Name ____ Date ___________ Period _____

Writing the Inverse of a Linear Function

TEACHER KEY | Algebra II | Unit 1 | Day 11 | TEK A2.2.B

Objective: I can write the inverse of a linear function algebraically by swapping x and y and solving for y, then check my answer by graphing or with a table.

Warm-Up — You Already Know How to Do This

In Algebra I you solved literal equations for a variable. That is the whole skill today.

Solve $y = 3x + 6$ for x: $x =$ **$(y - 6)/3$**

The Three Steps

Step 1. Replace $f(x)$ with **y**.

Step 2. **Swap x and y**.

Step 3. Solve for y, then write it as **$f^{-1}(x)$**.

Part 1 — Worked Examples

Example 1. $f(x) = 2x + 3$

Swap: **$x = 2y + 3$** Solve: **$y = (x - 3)/2$**

So $f^{-1}(x) =$ **$(x - 3)/2$**

Example 2. $f(x) = 3x - 5$

Swap: **$x = 3y - 5$** Solve: **$y = (x + 5)/3$**

So $f^{-1}(x) =$ **$(1/3)x + 5/3$**

Part 2 — You Try

1. $f(x) = x + 7$

   $f^{-1}(x) =$ **$x - 7$**

2. $f(x) = 4x$

   $f^{-1}(x) =$ **$x/4$**

3. $f(x) = 5x - 10$

   $f^{-1}(x) =$ **$(x + 10)/5$**

4. $f(x) = -2x + 8$

   $f^{-1}(x) =$ **$(8 - x)/2$**

Part 3 — Check It Two Ways

Use $f(x) = 2x + 3$ and $f^{-1}(x) = (x - 3)/2$ from Example 1.

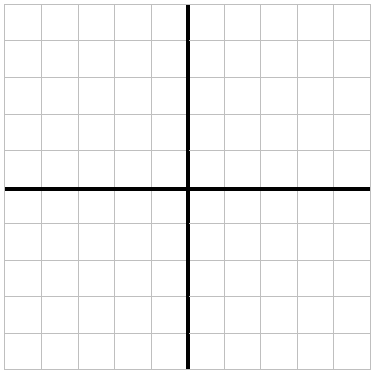
Check 1: the tables should trade columns.

x	-1	0	1	2
f(x)	1	3	5	7

x	1	3	5	7
$f^{-1}(x)$	-1	0	1	2

Check 2: plot both. They must mirror across y = x.

Graph f and f^{-1}, plus the line y = x.



Record it.

Slope of f: **2**

Slope of f^{-1}: **1/2**

What is the relationship?

the slopes are reciprocals

y-intercept of f is 3. Of f^{-1}?

-3/2

Part 4 — Watch Out

A student says the inverse of f(x) = x + 4 is $f^{-1}(x) = 1/(x + 4)$. What went wrong?

they used the reciprocal instead of undoing the function; the answer is x - 4

Which linear function is its own inverse? **f(x) = x (also f(x) = -x)**

What is the inverse of a horizontal line, like f(x) = 3? **not a function - every input maps to 3**

Part 5 — In Context

To convert Celsius to Fahrenheit: F(c) = (9/5)c + 32.

Find the inverse: $F^{-1}(x)$ = **(5/9)(x - 32)**

What does the inverse do in words? **converts Fahrenheit back to Celsius**

Practice

Write each inverse. Show the swap-and-solve steps.

1. f(x) = (1/2)x - 6

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$f^{-1}(x)$ = **2x + 12**

2. f(x) = -x + 9

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$f^{-1}(x)$ = **9 - x**

Exit Ticket

Write the inverse of f(x) = 6x - 12, then verify it with one input.

$f^{-1}(x)$ = **(x + 12)/6**

Check: f(3) = **6** and f^{-1}(**6**) = **3**